## MILITARY SPECIFICATION

# GAGES, PLUG AND RING, THREAD

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

### 1. SCOPE

1.1 This specification covers manufacture, inspection, and preparation for delivery requirements for Unified Standard and American National Standard thread plug, truncated thread setting plug, and thread ring gages listed in Military Standards 114, 115, 116, 117, 273, and 274 and for Unified Standard and American National Standard thread plug, truncated thread setting plug, and thread ring gages of special diameter-pitch combinations made to American Gage Design Standard specifications (AGD).

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

**SPECIFICATIONS**

FEDERAL

NN–P–515 — Plywood, Container Grade.

PPP–T–76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant.

PPP–B–585 — Boxes; Wood, Wirebound.

PPP–B–591 — Boxes, Fiberboard, Wood-cleated.

PPP–B–601 — Boxes, Wood, Cleated Plywood.

PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636 — Boxes, Fiber.

UU–P–268 — Paper, Kraft, Wrapping.

MILITARY

MIL–P–116 — Preservation, Methods of.

MIL–B–121 — Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL–S–45176 — Specimens, Comparison, Surface Roughness (Flat) with Carrying Case.

MIL–I–45177 — Instruments, Tracer, Surface Roughness.

MIL–G–45653 — Gages, Cylindrical Plug and Ring, Plain.

FSC 5220

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## STANDARDS

### MILITARY

MIL-STD-10 — Surface Roughness Waviness and Lay.

MIL-STD-114 — Gages, Plug, Thread Go (Class X) for Unified & American National Standard Internal Threads.

MIL-STD-115 — Gages, Plug, Thread Not Go for Unified & American National Standard Internal Threads.

MIL-STD-116 — Gages, Ring, Thread, Go (Class X) and Related Thread Setting Plug Gages, Go and Not Go Plain Plug Minor Diameter Acceptance Check Gages for Unified & American National Standard External Threads.

MIL-STD-117 — Gages, Ring, Thread, Not Go and Related Thread Setting Plug Gages, Go and Not Go Plain Plug Minor Diameter Acceptance Check Gages for Unified and American National Standard External Threads.

MIL-STD-120 — Gage Inspection.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-273 — Gages, Plug, Thread Setting, Class W, for Go Gages Unified Standard Classes 2A and 3A and American National Standard Class 3 External Threads.

MIL-STD-274 — Gages, Plug, Thread Setting, Class W, for Not Go Gages Unified Standard Classes 2A and 3A and American National Standard Class 3 External Threads.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other Publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

### U. S. DEPARTMENT OF COMMERCE

Commercial Standard CS8, Gage Blanks Supplement to Commercial Standard CS8.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

### CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

### AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification.

No. A-3 and No. 13.

(Application for copies of the above publication should be addressed to the American Trucking Associations, Inc. Traffic Department, 1424 Sixteenth Street, N. W., Washington 6, D. C.)



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U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations.

(Application for copies should be obtained from any first class U. S. Post Office.)

## 3. REQUIREMENTS

### 3.1 Gaging members.

3.1.1 *All gages.*

3.1.1.1 Gaging members shall be of American Gage Design Standard (AGD) as specified in Commercial Standard CS8 and the Supplement to Commercial Standard CS8.

3.1.1.1.1 Unless otherwise specified, the gaging members shall be of a high quality, electric furnace alloy steel, alloy tool steel, or carbon tool steel and shall have a minimum hardness equivalent to C60 Rockwell.

3.1.1.1.2 *Finish of flank of threads.* All gages shall be precision lapped to remove amorphous or smear metal left after finish grinding. Thick end threads (both ends of gage) shall removed by lapping.

*Note*: Elimination of this condition by undercutting (thin end threads) or by convoluting beyond the first full thread is not permitted.

3.1.1.1.2.1 Numerical values denoted on documents or drawings for gaging and other surfaces of the gages are Roughness Height Ratings as described in Standard MIL–STD–10.

3.1.1.1.3 *Sharp edges.* Sharp edges shall not be permitted except at the intersection of gaging surfaces.

3.1.1.1.4 *Stability.* The material selected and the appropriate artificial seasoning employed shall insure that the distortion or dimensional growth is held to a minimum and shall remain within the limits of the size specified.

3.1.1.1.5 *Gage tolerances.* Gage tolerances on lead, angle, and pitch diameter are not cumulative; that is, the tolerance on any one element shall not be exceeded even though the errors in the other two elements are smaller than the respective tolerances.

3.1.1.1.6 *Marking.* All marking shall be permanent and legible and of suitable height commensurate with the size of the handle or gaging member. Fractional marking except for the number sizes, is permitted in lieu of the decimal method of designation. If the gage size is marked in decimals, when marking the thread designation, there need only be sufficient decimal places to define the size. Example: .190, .5, .625, 1.0, 2.5, 3.75 etc.

3.1.2 *Thread plug gages and truncated thread setting plug gages.*

3.1.2.1 Gaging members up to and including 1.510 inches in diameter shall be finished from AGD thread blanks of the taperlock design with precision centers. Gaging members of thread size .150 inch or below (No. 0 to 6 incl.) when made with male centers shall have the male precision centers left on the gaging members. Gaging members above 1.510 inches in diameter shall be finished from AGD thread blanks of the trilock design.

3.1.2.2 Go thread plug gaging members shall be provided with a chip groove as specified on related applicable documents.

3.1.2.3 *Partial thread.* The partial thread at both ends of the gaging member shall be removed to a blunt start to avoid feather edges. Not more than one complete turn of the thread shall be removed to the point where the full thread form is obtained. On thread gaging members of 28 threads per inch and finer, a 60° chamfer from the axis of the gage is permitted in lieu of removal of the partial thread (see 3.1.1.1.2). Gages



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that have male precision centers left on the gaging members perform the same function as the 60° chamfer (see 3.1.2.1). On go thread plug gaging members with a chip groove, the partial thread preceding the chip groove shall be completely removed to a depth below the root of the thread. The truncated portion of truncated thread setting plug gaging members where it meets the full form portion shall be convoluted (blunt start) with the helix angle on gages of 28 threads per inch and coarser to avoid a feather edge.

3.1.2.4 The thread flanks for go thread plug and truncated thread setting plug gaging members shall be straight to at least the minor diameter (p/4) of the gaging member thread form. The minor diameter shall be cleared beyond a p/8 width of flat either by an extension of the sides of the thread toward a sharp "V" or by an undercut no wider than a width obtained from a p/8 maximum width either side of the centerline of the thread space. The root clearance for truncated thread setting plug gages for go thread ring gaging members must be such that the minimum minor diameter of the thread ring gage is cleared after the go thread ring gaging member has been properly set. The thread flanks for not go thread gaging members shall be straight and if undercut shall at least extend to a root clearance of 5p/16 max. The clearance form is optional provided the depth is sufficient to clear the minimum minor diameter of the component internal thread. The clearance form should be approximately central with the flanks of the thread.

3.1.2.5 The tolerance on lead for go and not go thread plug gaging members is the maximum allowable amount, measured in an axial direction, that any point on the helical path may deviate from its true position with respect to any other point within a distance equal to the length of the standard blank.

3.1.2.6 The tolerance on lead for truncated thread setting plug gaging members is the maximum allowable amount, measured in an axial direction, that any point on the helical path may deviate from its true position with respect to any other point within a distance equal to the width of the mating ring.

3.1.2.7 *Taper of pitch diameter.* The maximum permissible taper of the pitch diameter for truncated thread setting plug gages over the entire length of the gaging member shall be within the following limits: For thread size up to and including 4 inches the taper shall not exceed .0001 inch, except that for threads coarser than 16 threads per inch the taper shall not exceed .00015 inch. For thread sizes larger than 4 inches the taper shall not exceed .0002 inch. The permissible taper shall be back taper (largest diameter at entering end) and shall be within the pitch diameter limits specified on the applicable documents.

3.1.3 *Thread ring gages.*

3.1.3.1 *Partial thread.* The partial thread at both ends of the gage shall be removed to a blunt start to avoid feather edges. Not more than one complete turn of the thread shall be removed to the point where the full thread form is obtained. On thread ring gages of 20 threads per inch and finer or a nominal size of ½ inch or smaller a 60° chamfer from the axis of the gage is permitted in lieu of the removal of the partial thread (see 3.1.1.1.2).

3.1.3.2 *Thread flanks.*

3.1.3.2.1 The thread flanks for go thread ring gages shall be straight and at least extend to a root clearance of p/8 max. The clearance form is optional and should be approximately central with the flanks. The root clearance must be such that the maximum major diameter of the full form portion



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of the thread setting plug gage is cleared after the go thread ring gage has been properly set.

3.1.3.2.2 The thread flanks for not go thread ring gages shall be straight and at least extend to a root clearance of 5p/16 max. The clearance form is optional and may be an extension of the sides of the thread toward a sharp V except that the thread root shall be relieved (undercut root relief) for gages that have pitches coarser than 20 threads per inch. The clearance form or extension toward an absolute sharp V must have sufficient depth to clear the maximum major diameter of the full portion of the truncated thread setting plug gage or the maximum major diameter of the external thread, whichever is greater, after the not go thread ring gage has been properly set.

3.1.3.3 The limits of the minor diameter shall be inspected after the gaging member has been set to the applicable truncated thread setting plug gaging member.

3.1.3.4 The effective minor diameter shall be inspected. To be acceptable the actual (measured) minimum effective minor diameter shall not be less than the basic minor diameter minus the sum of the gagemaker's tolerance for the pitch and minor diameters respectively for go thread ring gages and shall not be less that basic minor diameter minus two times the sun of the gagemaker's tolerance for the pitch and minor diameters respectively for not go thread ring gages.

3.1.3.5 The tolerance on lead for go and not go thread ring gages is the maximum allowable amount, measured in an axial direction, that any point on the helical path may deviate from its true position with respect to any other point within a distance equal to the length of the standard blank.

3.1.3.6 *Plain plug acceptance check gages for minor diameter.* Plain plug acceptance check gages in accordance with the requirements of MIL–G–45653 shall be provided when specified for gaging members with basic size of minor diameter below .375 inch.

**3.2 Handles for thread plug gages and truncated thread setting plug gages.**

3.2.1 Handles shall be of American Gage Design Standard (AGD) in accordance with Commercial Standard CS8 and the Supplement to Commercial Standard CS8.

3.2.1.1 All gage handles shall be single end type.

3.2.2 Unless otherwise specified gage handles shall be made of aluminum alloy.

3.2.3 Taperlock design gage handles may be tapered on one end only and the drift hole or slot may be omitted.

3.2.4 Trilock design gage handles shall be provided with steel screws of type No. 2 as illustrated in the 1955 Supplement to Commercial Standard CS8.

3.2.4.1 The steel screws shall be free from burrs and torn threads.

3.2.5 The external surfaces of go thread plug gage handles shall be anodized with a permanent green color having a bright finish. The external surfaces of not go thread plug gage handles shall be anodized with a permanent red color having a bright finish. The external surfaces of truncated thread setting plug gage handles shall be either rough anodized or highly polished (not anodized) maintaining the natural aluminum color.

3.2.5.1 The annular groove for not go thread plug gage handles as shown in Commercial Standard CS8 is not required.

3.2.6 When specified, thread plug gaging members and acceptance check gaging members may be procured without handles. Gaging members shall be marked with the re-

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quired identifying data specified in related applicable documents as follows:

Wire Type — The gaging member shall be identified with a tag or label having the required pertinent data printed thereon. The tag or label shall be such that it can be removed without damage to the gaging member.

Taperlock Type — Marked by a steel stamp or electric etching on the front end of the gaging member or if space does not permit, marked on the taper portion of the gaging member.

*Note*: Marking shall not impede assembly of gaging member and handle.

Trilock Type — Marked by a steel stamp or electric etching on either side of the gaging member.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Contractor inspection.** Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

**4.1.1** Unless otherwise specified, the supplier will make such inspections as are necessary of each gage by use of one hundred percent inspection in conjunction with the methods and procedures described in Standard MIL–STD–120 to determine that the gages are in accordance with the requirements of this specification and applicable documents.

**4.2 Hardness test.** Hardness tests shall be made on standard type hardness testers and shall be made as near to the actual gage wear surface as practicable. Thread gages shall be tested on the end of the gage adjacent to the threads. The actual threads shall have an equivalent hardness.

**4.3 Surface roughness of gaging surfaces.** Surface roughness shall be determined by an instrument conforming to the requirements of specification MIL–I–45177. Areas where measurement of surface roughness is difficult or impracticable determination of the surface roughness height value shall be made by visual comparison with type I specimens conforming to Specification MIL–S–45176.

**4.3.1** Gaging surfaces shall be inspected for seams, cracks, nicks, scratches, rust spots and other defects which may materially effect the use, durability, or stability of the gage.

**4.4 Gaging members and handles.** Taperlock gaging members and handles shall be inspected to insure correct taper and size.

**4.5 Stability test.** When deemed necessary, gages will be subjected to a deep freeze treatment to determine whether or not they have been properly stabilized. Gages shall remain in tolerance after treatment.

**4.6 Lead error.**

**4.6.1** All thread gaging members will be inspected for total lead error (cumulative effect of progressive, erratic, and drunkenness errors). Measurements will be taken from a fixed reference point located at the start of the first full thread to a sufficient number of positions along the entire helix to detect all types of lead deviations which may be present. The amount that these positions deviate from their basic (theoretical) positions will be recorded with due respect to sign.

**4.6.1.1** The total spread between the deviation readings taken between any two threads



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not further apart than the length of the standard blank shall not exceed the specified tolerance on lead for thread plug and thread ring gaging members.

**4.6.1.2** The total spread between the deviation readings taken over a length of threads equal to the width of the mating ring shall not exceed the specified tolerance on lead for truncated thread setting plug gaging members.

## 5. PREPARATION FOR DELIVERY

**5.1 Preservation and packaging.** Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The gages shall be cleaned in accordance with process C–1 of Specification MIL–P–116.

5.1.1.2 *Drying.* The gages shall be dried in accordance with any applicable drying procedure specified in Specification MIL–P–116.

5.1.1.3 *Preservative application.* The gages shall be preserved with type P–10 preservative as specified in Specification MIL–P–116.

5.1.1.4 *Unit packaging.* Each gage shall be packaged by method IA–15 as specified in Specification MIL–P–116 and herein. The gage shall be wrapped with type II, grade A, class 2, barrier material conforming to Specification MIL–B–121 and secured with tape conforming to Specification PPP–T–76. The MIL–B–121 barrier material shall be overwrapped with Kraft paper conforming to Specification UU–P–268.

5.1.2 *Level C.* The gages shall be preserved and packaged in accordance with the manufacturer's commercial practice.

**5.2 Packing.** Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The gages (gages of the same Federal stock N.) shall be packed in a close-fitting wire bound wood box, wood cleated fiberboard box, wood cleated-plywood box, nail wood box, or a fiberboard box conforming to Specification PPP–B–585, class 3, PPP–B–591 (overseas type), PPP–B–601, style A or B using plywood conforming to Specification NN–P–515, type II or III, class 2, PPP–B–621, class 2, and PPP–B–636, class 2 of 3, respectively. Closure and strapping requirements shall be in accordance with the applicable box specification or appendix thereto. The gross weight for wood or wood-cleated boxes shall be approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specifications.

5.2.2 *Level B.* The gages (gages of the same Federal stock N.) shall be packed in close-fitting wire bound wood box, wood-cleated fiberboard box, wood cleated-plywood box, nail wood box or a fiberboard box conforming to Specification PPP–B–585, class I, PPP–B–591 (domestic type), PPP–B–601, (domestic type), PPP–B–621, class I, and PPP–B–636, class I, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall be approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The gages shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classifications, Motor Freight Regulations or Parcel Post Regulations, whichever is applicable.



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**5.3 Marking.** Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL–STD–129.

## 6. NOTES

**6.1 Intended use.** Gages described herein are to be used for acceptance inspection by Government facilities.

**6.2 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Number of gages required and federal stock number or gage drawing number.

(c) Place of delivery for acceptance inspection.

(d) Any required acceptance check gages (see 3.1.3.6).

(e) Level of preservation, packaging, and packing required.

(f) Material if different from 3.1.1.1.1 or 3.2.2.

(g) Procurement of gaging members only (see 3.2.6).

**Notice:** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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